

Sean Luka Girgis

Senior Performance Engineer

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Professional Summary

Senior Performance Engineer with 20+ years of enterprise IT experience across performance testing, APM/observability, Java/J2EE diagnostics, telemetry analytics, capacity forecasting, and production troubleshooting. Strong fit for LoadRunner/JMeter performance engineering, Dynatrace/AppDynamics monitoring, AWS-aware cloud performance analysis, and cross-functional performance governance. Known for translating complex performance metrics into actionable tuning recommendations, executive-ready reports, dashboards, baselines, and capacity-risk insights for mission-critical systems.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Led enterprise capacity and performance analytics for banking infrastructure using telemetry from 6,000+ endpoints, translating CPU, memory, utilization, and application-performance signals into risk forecasts and executive reporting.
- Built automated Python/Pandas ETL pipelines ingesting monitoring and capacity data from BMC TrueSight/TSCO, CA APM, and AppDynamics, replacing manual processes with repeatable reporting workflows.
- Integrated disparate monitoring feeds into unified Oracle reporting datasets with validation, reconciliation, trend analysis, and data quality controls for performance and capacity decisions.
- Developed Prophet and scikit-learn forecasting models to predict capacity bottlenecks months ahead and support proactive remediation before service-impacting events.
- Created dashboards and performance reports using Streamlit, matplotlib, seaborn, and plotly to present capacity posture, utilization trends, bottlenecks, and optimization recommendations to senior stakeholders.
- Partnered with application, infrastructure, and operations teams to investigate performance issues, identify systemic bottlenecks, and document actionable tuning and remediation plans.
- Supported cloud modernization patterns using AWS S3, Glue, Redshift, EC2/ECS, and Python ETL pipelines, including cost/performance tradeoff analysis for forecasting workloads.
- Produced operational runbooks, repeatable reporting processes, and stakeholder-ready summaries supporting governance, SLA awareness, and continuous improvement.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, strengthening production observability for web and mobile platforms.
- Led the FAST project, mining real-user and synthetic performance metrics to identify bottlenecks, optimization opportunities, and recommendations adopted by engineering teams.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior with APM telemetry and support performance-driven release decisions.

- Implemented End-User Monitoring and end-to-end transaction tracing across web and mobile applications to improve visibility into user experience and application behavior.
- Supported Dynatrace upgrade and TLS 1.2 security cutover with no production monitoring regression, balancing performance visibility, security, and operational continuity.
- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, performance risks, and operational data requirements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large enterprise environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules that improved application-performance visibility and became standards across client environments.
- Partnered with J2EE/WebLogic performance teams to diagnose bottlenecks using APM telemetry, SQL/Oracle data, JVM behavior, thread activity, and infrastructure signals.
- Built Perl and Korn Shell extraction scripts to automate performance reporting and reduce manual weekly export work.
- Provided sizing recommendations, monitoring onboarding, golden-image agent deployment guidance, client training, and technical documentation for production APM operations.
- Led CA APM upgrade activities and cross-functional coordination with zero unplanned downtime across monitored enterprise systems.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits and resource bottlenecks across JDBC connections, threads, heap memory, CPU, and garbage collection behavior.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts for real-time visibility during performance test execution.
- Used HP Performance Center / LoadRunner environment exposure to support performance testing, bottleneck diagnosis, and release-readiness analysis.
- Documented performance baselines and regression thresholds used by engineering teams for release-gate decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI/OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness and performance behavior across large transaction datasets.
- Automated schema conversion and data validation workflows to protect data integrity and reduce migration risk during phased cutover.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability telecom billing interfaces using C, C++, Pro*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and production Unix environments.
- Delivered performance enhancements in billing processes, reducing memory usage, improving throughput, and improving database performance through SQL and sequence optimization.

- Maintained reporting, invoicing, settlement, and integration components across full billing lifecycle systems for large-scale telecom operations.
- Automated WebLogic/WebSphere administration and deployment support with Korn Shell scripting, reducing manual operational effort.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

HorizonScale — AI Capacity Forecasting Engine

Built a capacity forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform enterprise telemetry into bottleneck forecasts, dashboards, and operational performance insights.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Telemetry Analytics, Capacity Forecasting

FAST Project

Mined real-user and synthetic monitoring data from Dynatrace and Gomez Synthetic Monitoring to identify performance bottlenecks and optimization opportunities for critical revenue-generating systems.

Technologies: Dynatrace, Gomez Synthetic Monitoring, EUM/RUM, Transaction Tracing, Performance Analytics

Serverless Lakehouse Platform (AWS)

Designed AWS data platform patterns using S3, Glue, Athena, Redshift, PySpark, Parquet, and compression strategies to support cloud-scale ingestion, transformation, analytics, and cost-aware processing.

Technologies: AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet

AI-Powered Job Search Pipeline

Built an agentic Python automation pipeline with duplicate detection, scoring, JSON artifact generation, quality gates, and tracking, using FAISS, sentence-transformers, LLM APIs, and structured output contracts.

Technologies: Python, FAISS, Sentence Transformers, LLM APIs, RAG, JSON Contracts